

SINDy on the Four Moreton-Bay Tide Gauges

A side exercise: discovering a forced linear model from gauge data

PINN course — Unit 1 data, Unit 3 (SINDy) method

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The key idea

SINDy (Sparse Identification of Nonlinear Dynamics) takes a measured trajectory $\mathbf{x}(t) \in \mathbb{R}^n$, estimates its time derivative $\dot{\mathbf{x}}(t)$, and looks for the *sparsest* combination of candidate functions that reproduces it:

$$\dot{\mathbf{x}} = \Theta(\mathbf{x})\Xi, \quad \Theta(\mathbf{x}) = [1 \ \mathbf{x} \ \mathbf{x}^{\otimes 2} \ \dots],$$

where each column of Θ is a candidate term and the columns of Ξ are made sparse by a thresholded least squares (STLSQ). The premise is that the *right* variables obey a *closed, autonomous* law with only a few active terms.

Here we test that premise on real course data: the **four tide gauges** of Unit 1. The state is $\mathbf{x}(t) = (\eta_{G1}, \eta_{G2}, \eta_{G3}, \eta_{G4})$, the noise-free (**gauges_clean**) water-level anomalies, sampled at $\Delta t = 12$ s over 10 h ($N = 3001$ points). The twist: the bay is a *linearised shallow-water system*, and it is **forced** by an external river surge $\psi(t)$ at the Brisbane River mouth — a pulse peaking at $\psi \approx 0.45$ around $t \approx 2$ h, then decaying. So the honest model class is a *forced linear* system,

$$\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{b}\psi(t)$$

and the central question becomes: *can SINDy see the forcing?*

What we ran

Two fits, both with a constant + linear library and column-normalised STLSQ (relative threshold $\lambda = 0.05$); derivatives by central differences:

- **(A) Autonomous** — library $[1, \eta_{G1\dots G4}]$ only: $\dot{\mathbf{x}} = A\mathbf{x}$.
- **(B) Controlled (SINDYc)** — append the known forcing $\psi(t)$ as an extra column: $\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{b}\psi$.

Each identified model is then integrated forward (RK4) from rest, $\mathbf{x}(0) = \mathbf{0}$, and compared to the gauge data.

The discovered system

The controlled fit returns a *stable, oscillatory* forced linear model (all entries $\times 10^4$ for readability):

$$10^4 A = \begin{pmatrix} +4.20 & +4.29 & -8.13 & -1.08 \\ +3.28 & +3.08 & -6.79 & -0.32 \\ +5.76 & +4.55 & -9.42 & -1.08 \\ +3.51 & -4.97 & +2.53 & -0.03 \end{pmatrix}, \quad 10^4 \mathbf{b} = \begin{pmatrix} +0.31 \\ +0.32 \\ -0.02 \\ +0.14 \end{pmatrix}.$$

Eigen-analysis of A . Two lightly-damped oscillatory pairs,

$$\lambda \approx (-5.4 \pm 9.0 i) \times 10^{-5} \text{ and } (-5.4 \pm 32 i) \times 10^{-5} \text{ s}^{-1},$$

i.e. resonant **periods** $\approx 19.3 \text{ h}$ and $\approx 5.4 \text{ h}$, each with an e -folding decay of $\approx 5 \text{ h}$. SINDy recovered the bay’s physical mode structure from four scalar timeseries, and $\max \text{Re}(\lambda) = -5.4 \times 10^{-5} < 0$, so the model is stable to integrate forward.

Forcing vector $\mathbf{b}$. The surge enters predominantly through $G1$ and $G2$ — the gauges nearest (and across from) the river mouth — and ≈ 0 at $G3$ (Tangalooma, in the Moreton-Island lee). The identified forcing direction is physically where you would expect it.

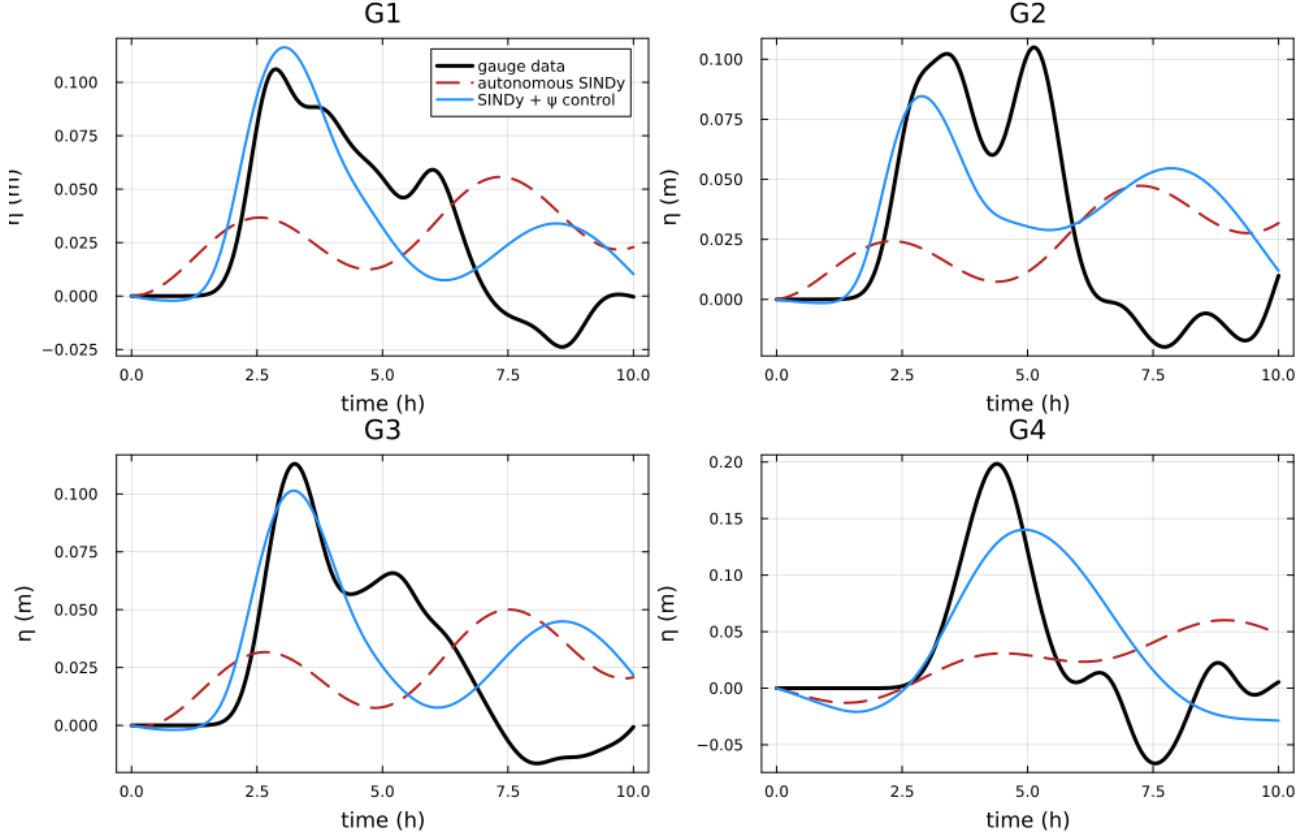


Figure 1: Each panel: gauge data (**black**) vs. the two identified models, integrated from rest. **Autonomous SINDy** (red dashed) has no forcing term, so from rest it never leaves rest — it cannot manufacture the surge. **SINDy + ψ control** (blue) captures the onset, timing and primary peak of the surge response (near-exact at $G3$), driven by the recovered $\mathbf{b}\psi(t)$. It misses the secondary peaks and late ringing — see the closure caveat below.

Three lessons

1. **Forced systems need a control input.** Autonomous SINDy fails outright: per-gauge reconstruction error $\approx 90\text{--}100\%$. With no forcing term, a model started from rest stays at rest. Adding the known $\psi(t)$ (SINDy_c) is what makes the surge appear — error drops to $\approx 57\text{--}80\%$ and the primary event is captured.
2. **SINDy can recover real physical structure.** The eigenvalues of the identified A are the bay’s

damped resonant modes (≈ 5.4 h, ≈ 19.3 h), and $\mathbf{b}$ localises the forcing to the river-side gauges — both physically meaningful, read straight off the coefficients.

3. **Closure matters — and four point-gauges are *not* closed.** The residual misfit (secondary peaks, late ringing) is not a tuning failure: the gauges are a *partial observation of a PDE field*, so the true gauge dynamics depend on the whole hidden bay state, and no four-state ODE can be exact. Closing the gap is the natural next step — **time-delay embeddings** (HAVOK) or more observables.

Why this is a good Unit-3 example

It teaches SINDy’s two central caveats at once on a real, multivariate, *forced* dataset — (i) autonomous vs. control inputs, and (ii) the need for a closed state — and it threads Unit 3 back to the Unit 1 surge inverse problem, where recovering exactly this $\psi(t)$ is the headline task.

Data: `units/unit_01/data/{gauges_clean, psi_truth}.csv`. **Method:** central-difference $\dot{\mathbf{x}}$, constant + linear library, column-normalised STLSQ. Exploratory; not in the repository.